Wildfire Suppression Quadcopter

Electric, Remote-Piloted, Vertical Takeoff and Landing Cargo Aircraft

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Capstone Senior Design

Scan QR code to view competition fly-off videos

Competition Scope

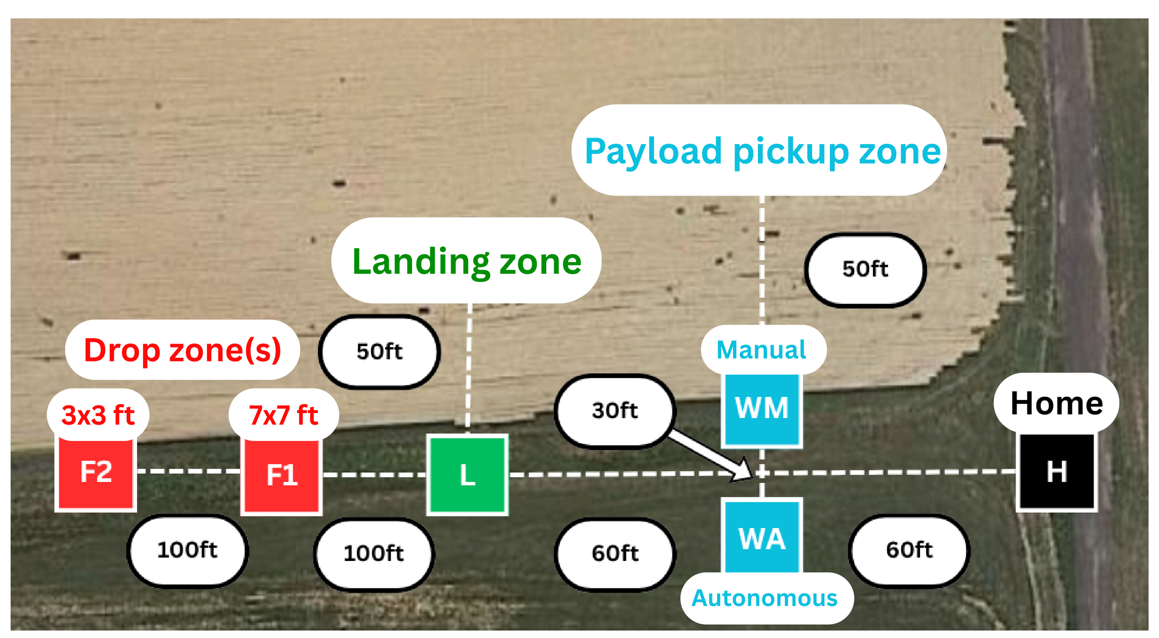
We are tasked with designing and building a remote-piloted eVTOL aircraft for the **Vertical Flight Society's DBVF Competition**.

Key Aircraft Parameters	
Takeoff Weight - 17.5/30 lb (Min/Max)	Cruise Endurance - 15 min
Size - 4 x 4 x 1 ft	Cruise Speed - 82 ft/s
Thrust to Weight Ratio - 3.5	Cruise Tilt - 75°

Missions (Autonomous flight earns bonus points):
FM1: Take off from H with a payload and land at L.
FM2: Take off from L and drop payload at F1 or F2.
FM3: Cycle between pickup at WA or WM and drop at F1 or F2 for the remaining time before landing at H.

10 min time limit

Points awarded for every accurate payload drop.

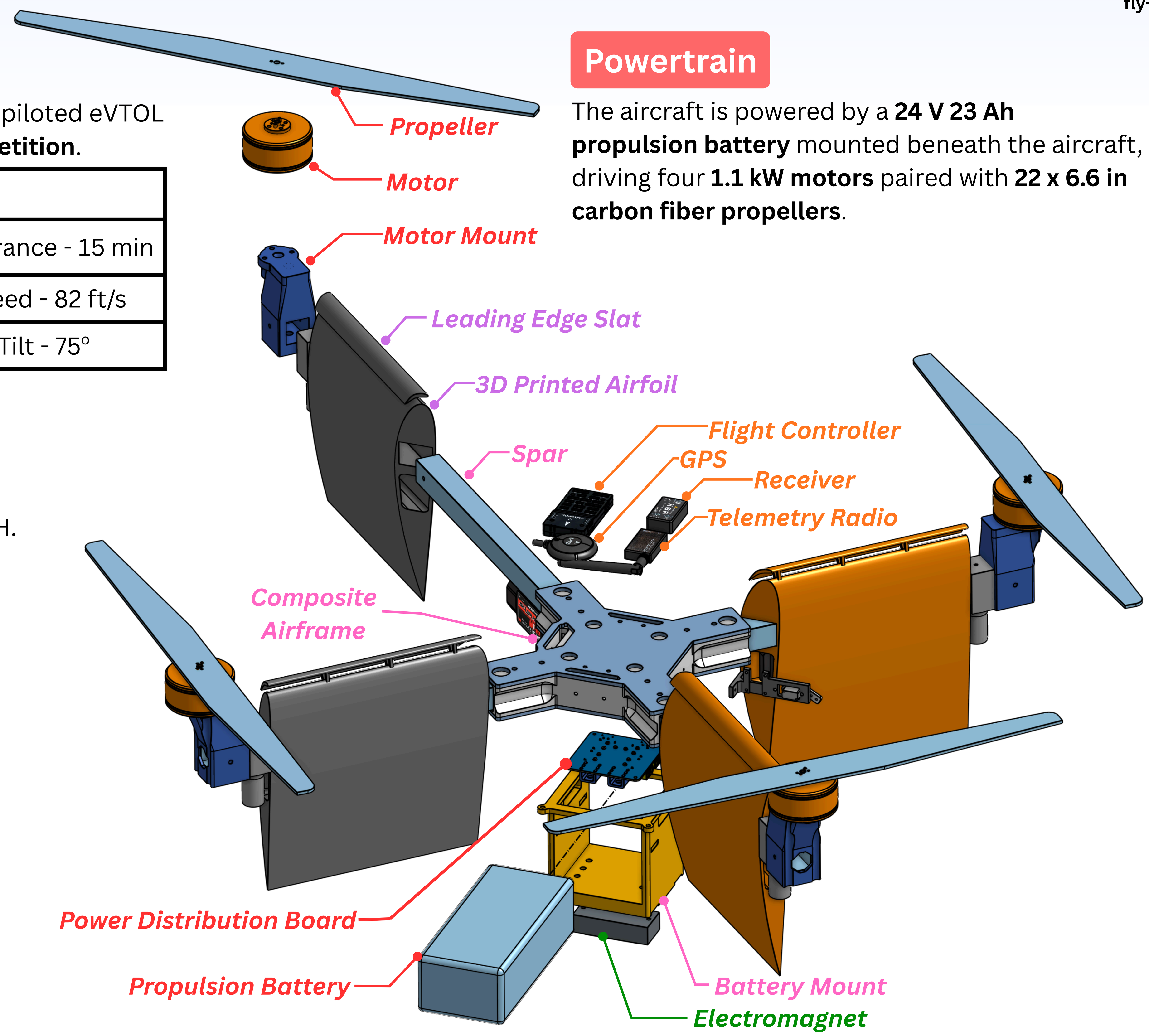


Competition course layout

Payload System

The aircraft uses an electromagnet to pick up and release **2.5 lb payloads** representing fire suppressant bags. Payloads must be released from **30 ft above the ground**.

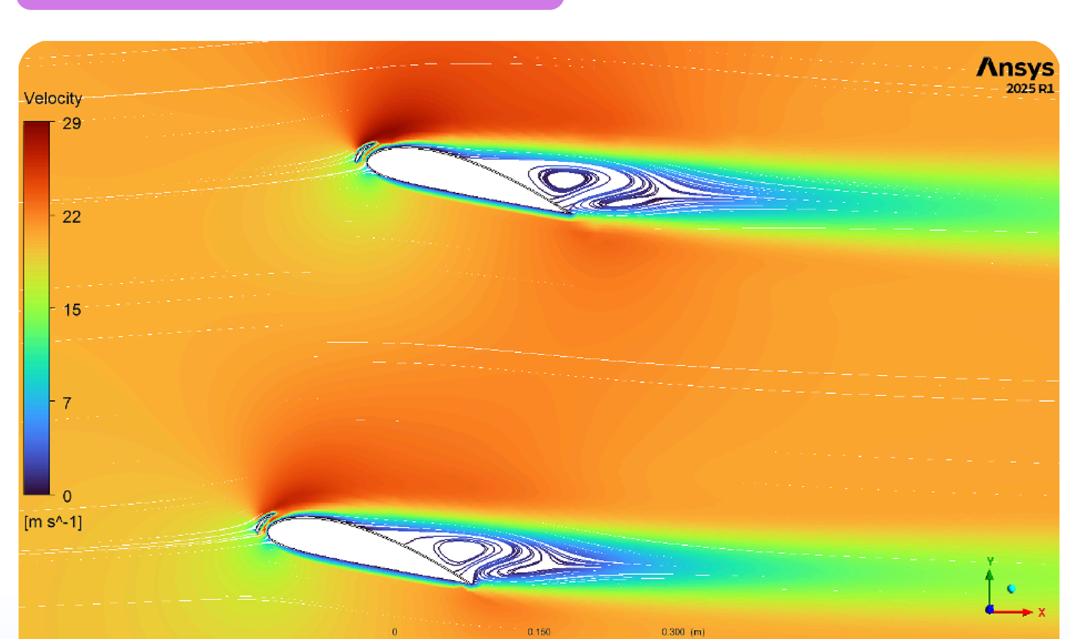
A **hanging electromagnet** design was chosen to minimize moving parts and allow pickups from above ground effect.



Powertrain

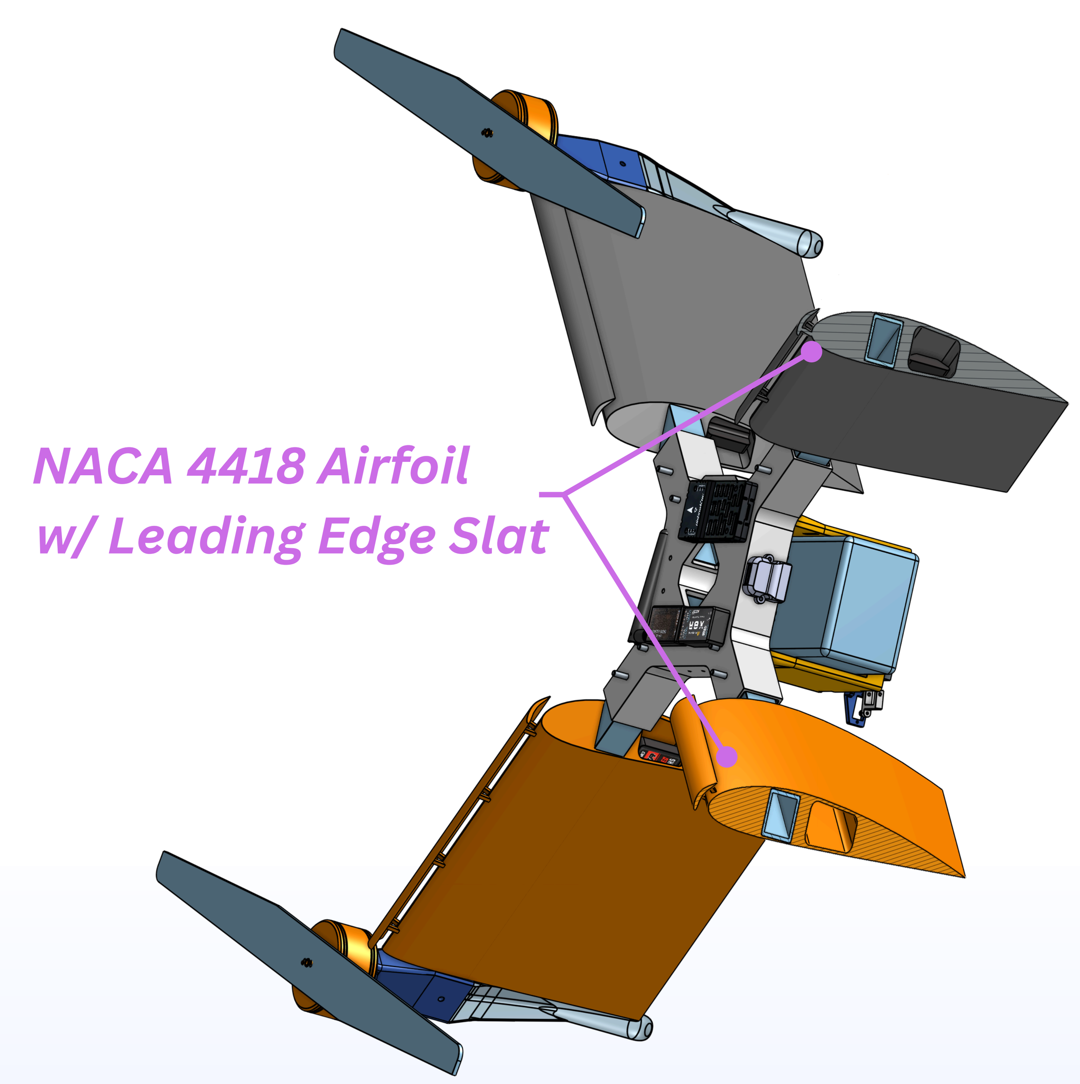
The aircraft is powered by a **24 V 23 Ah propulsion battery** mounted beneath the aircraft, driving four **1.1 kW motors** paired with **22 x 6.6 in carbon fiber propellers**.

Aerodynamics



Flow Around Airfoils at 15° AoA (75° Tilt)

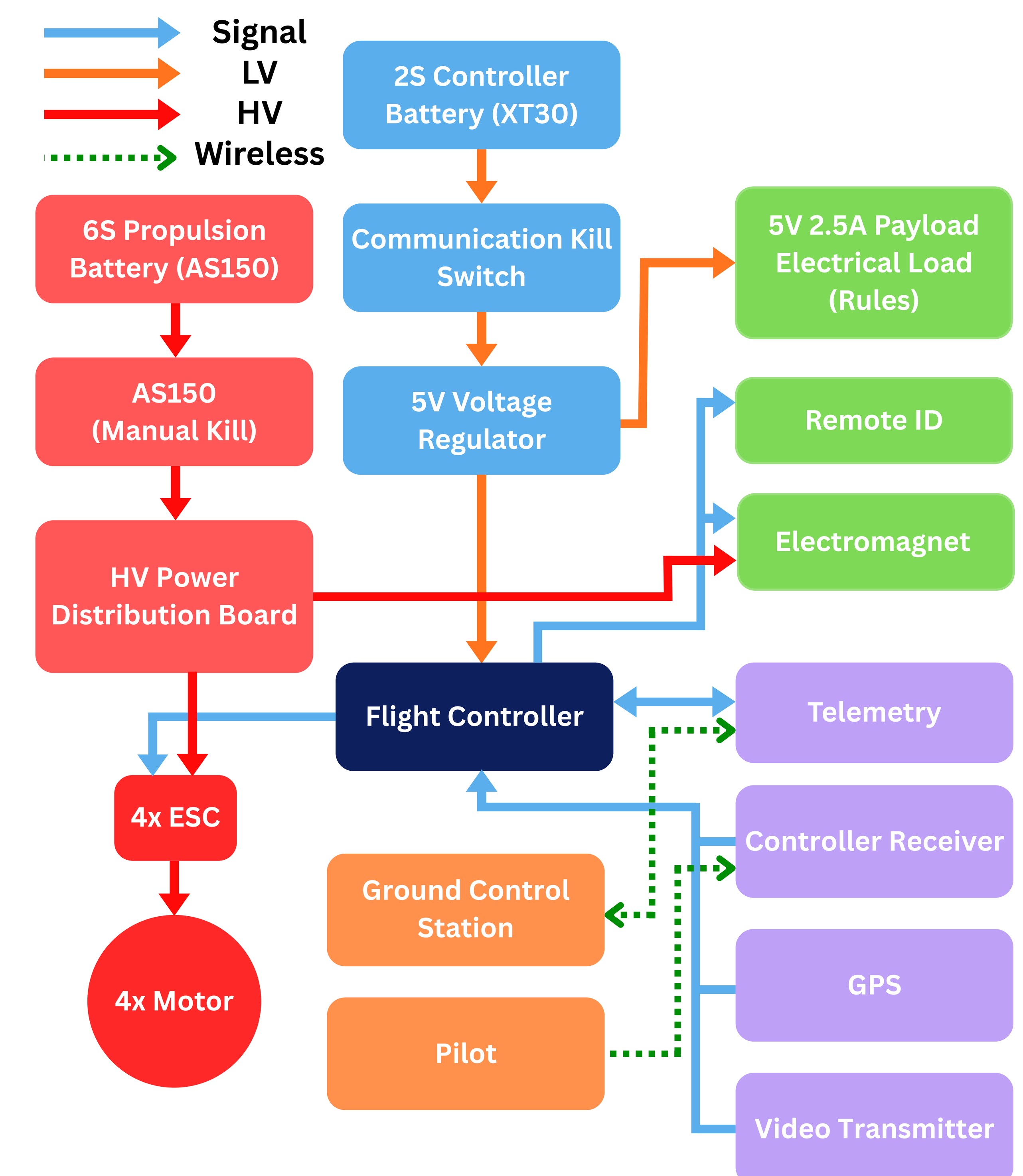
Carbon fiber spars incorporate NACA-4418 airfoils with leading-edge slats, resulting in **27% greater efficiency**.



Section View of Aircraft during Horizontal Cruise (75° tilt)

Electronics

The Holybro Pixhawk 6C flight controller fuses receiver input with **GPS** and **compass** data, then commands each motor's ESC and streams telemetry.



Competition Results

Placed 5th of 20 teams and earned the **2nd best technical report**, Cooper's best VFS finish to date. We scored **60.5%** of available points (**goal: 50%**). In our best attempt, we completed FM1 and FM2 autonomously and 5 FM3 pickups & drops in under 10 minutes.

Photo Credit: Tori Arcilesi

Acknowledgements: Thank you to Mike Giglia for helping us safely conduct tests